

*MA.6.AR.3.4***Benchmark****MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.**

Example: Gerald is trying to gain muscle and needs to consume more protein every day. If he has a protein shake that contains 32 grams and the entire shake is 340 grams, what percentage of the entire shake is protein? What is the ratio between grams of protein and grams of non-protein?

Benchmark Clarifications:

Clarification 1: Instruction includes the comparison of $\frac{\text{part}}{\text{whole}}$ to $\frac{\text{percent}}{100}$ in order to determine the percent, the part or the whole.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.1, MA.6.NSO.2.3
- MA.6.NSO.3.1, MA.6.NSO.3.5
- MA.6.AR.2.3

Terms from the K-12 Glossary

- Rate

Vertical Alignment**Previous Benchmarks**

- MA.5.NSO.2.1, MA.5.NSO.2.2
- MA.5.FR.1.1

Next Benchmarks

- MA.7.AR.3.1

Purpose and Instructional Strategies

In previous courses, students worked with creating equivalent fractions, as well as operations with whole numbers and fractions. In grade 6 accelerated, students use ratio relationships to solve problems involving percentages. This extends in grade 6 accelerated, where students will solve multi-step problems and proportional relationships involving percentages. In future courses, students will solve multi-step real-world percent problems by applying previous understanding of percentages and ratios.

- Students should understand that percent (%) represents a part to whole relationship.
- Instruction includes the connection to ratio relationships to determine the part, the whole or the percentage (*MTR.5.1*).
 - For example, when determining how much 40% is of 24, students should compare the ratio $\frac{x}{24}$ to the ratio $\frac{40}{100}$.
- Instruction does not include the use of proportions or cross multiplication to solve problems involving percentages.

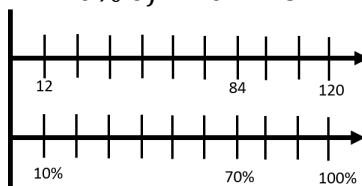
- Instruction includes the use of models to represent percentages such as bar models, number lines or ratio tables to help visually represent the relationship (*MTR.2.1*).
 - Bar Models

$$70\% \text{ of } 120 = 84$$

10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
12	12	12	12	12	12	12	12	12	12

- Number Lines

$$70\% \text{ of } 120 = 84$$



- Ratio Tables

$$70\% \text{ of } 120 = 84$$

percent	100%	50%	10%	70%
total	120	60	12	84

Common Misconceptions or Errors

- Students may not understand the difference between an additive relationship and a multiplicative relationship.
- Students may incorrectly set up ratios because of a misunderstanding of the part and the whole addressed in the situation.
- Students may not recognize simplified forms of ratios in order to find equivalent ratios to determine the percentage, the whole or the part.

Strategies to Support Tiered Instruction

- Instruction includes finding an equivalent unit rate (either part or whole), then multiplying to find the desired equivalent ratio.
 - For example: Steve wants to determine how much a 15% tip is if the bill is \$80.00.
Use a visual representation to show 80 represents 100%

Part	80
Percent	100

Divide the ratio by 80

Part	80	1
Percent	100	1.25

Multiply the resulting ratio by 15

Part	80	1	12
Percent	100	1.25	15

Using the equivalent ratios, 12 is 15% of 80, so the tip is \$12.00.

Instructional Tasks

Instructional Task 1 (MTR.4.1, MTR.6.1, MTR.7.1)

Carlos predicts that his math homework will take him 60 of the total of 75 minutes he has available for homework tonight.

Part A. At this rate, how many minutes would Carlos spend on math homework out of a total of 100 available minutes?

Part B. What percentage of the available homework time does Carlos predict he will spend doing math? Explain how the answer to this question is related to the answer in Part A.

Instructional Items

Instructional Item 1

Find the percent equivalent to $\frac{60}{115}$. Round to the nearest tenth percent.

Instructional Item 2

15% of 80 is what value?

Instructional Item 3

Sami is keeping track of the amount of salt she consumes each day. According to the label on her macaroni and cheese box, one serving contains 470 mg of sodium (salt). If 470 mg is 20% of the recommended daily amount, how many milligrams of sodium are recommended for the whole day?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*